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Research Article

3-indoleacetonitrile attenuates biofilm formation and enhances sensitivity to imipenem in *Acinetobacter baumannii*

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One sentence summary: 3-indoleacetonitrile displayed antibiofilm effect via inhibition of quorum sensing and enhanced the efficacy of imipenem against *A. baumannii*.

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Abstract

Acinetobacter baumannii poses a global danger due to its ability to resist most of the currently available antimicrobial agents. Furthermore, the rise of carbapenem-resistant *A. baumannii* isolates has limited the treatment options available. In the present study, plant auxin 3-indoleacetonitrile (3IAN) was found to inhibit biofilm formation and motility of *A. baumannii* at sublethal concentration. Mechanistically, 3IAN inhibited the synthesis of the quorum sensing signal 3-OH-C12-HSL by downregulating the expression of the *abaI* autoinducer synthase gene. 3IAN was found to reduce the minimum inhibitory concentration of *A. baumannii* ATCC 17978 against imipenem, ofloxacin, ciprofloxacin, tobramycin, and levofloxacin, and significantly decreased persistence against imipenem. Inhibition of efflux pumps by downregulating genes expression may be responsible for enhanced sensitivity and low persistence. 3IAN reduced the resistance to imipenem in carbapenem-resistant *A. baumannii* isolates by downregulating the expression of OXA β-lactamases ($bla_{oxa\text{-}51}$ and $bla_{oxa\text{-}23}$), outer membrane protein *carO*, and transporter protein *adeB*. These findings demonstrate the therapeutic potential of 3IAN, which could be explored as an adjuvant with antibiotics for controlling *A. baumannii* infections.

Keywords: *Acinetobacter baumannii*, biofilm, 3-indoleacetonitrile, resistance, imipenem

Introduction

Acinetobacter baumannii is an opportunistic human pathogen associated with nosocomial infections, such as ventilator-associated pneumonia, bacteraemia, septicaemia, meningitis, endocarditis, wound infections, and urinary tract infections. It is responsible for significantly increased morbidity and mortality in intensive care units (ICUs; Jain et al. 2019, Ayoub Moubareck and Hammoudi Halat 2020). *A. baumannii* has developed resistance to antimicrobial agents either through intrinsic and acquired resistance (Vázquez-López et al. 2020, Kyriakidis et al. 2021). During the last two decades, carbapenems have been increasingly used to treat MDR *A. baumannii* infections, resulting in a global prevalence of carbapenem-resistant *A. baumannii* (CRAB; Ayobami et al. 2019, Piperaki et al. 2019). CRAB has been classified as a global and critical priority pathogen by the World Health Organization (Tacconelli et al. 2018). Despite the fact that polymyxins and tigecycline are used as the last-resort drugs for CRAB infection, resistance to these antibiotics has risen in clinical *A. baumannii* isolates worldwide (Nowak et al. 2017, Şimşek and Demir 2020). CRAB has also been associated with coinfection and secondary infection in patients with COVID-19 (Rangel et al. 2021). A study from ICUs and wards with critically ill COVID-19 patients of 10 hospitals of Indian Council of Medical Research (ICMR) surveillance network identified *A. baumannii* as one of the most predominant pathogen responsible for secondary infections (Vijay et al. 2021). There is a positive correlation between multiple drug resistance and biofilm formation in *A. baumannii* (Zeighami et al. 2019, Uruén et al. 2020). Biofilm survival is often dependent on the formation of persister subpopulations, which are phenotypic variants that are tolerant to various stresses such as antibiotics, and are responsible for recurrence of infections (Carvalho et al. 2018). Thus new therapeutic strategies are urgently required to control *A. baumannii* infections.

3-indoleacetonitrile (3-indolylacetonitrile, 3IAN), an indole derivative, is a plant growth hormone auxin, produced by cruciferous vegetables (Brassica), such as broccoli, cabbage, and cauliflower (de Vos et al. 2008). Previously, 3IAN has been shown to decrease biofilm formation of *Escherichia coli* O157:H7 and *Pseudomonas aeruginosa* by inhibiting polymeric matrix production (Lee et al. 2011). In *Serratia marcescens,* 3IAN decreased biofilm formation, motility, and prodigiosin production by interfering with quorum sensing (QS; Sethupathy et al. 2020). Recently, 3IAN has been shown to exhibit antiviral activity against influenza A viruses both *in vitro* and *in vivo* (Zhao et al. 2021).

In the present study, 3IAN significantly reduced biofilm formation and motility in *A. baumannii* ATCC 17978 through the inhibition of QS. 3IAN enhanced sensitivity of antibiotics against *A. baumannii* ATCC 17978 and reduced persistence to imipenem. 3IAN acted as an efflux pump inhibitor by downregulating multiple efflux pumps. Furthermore, it enhanced susceptibility to imipenem in CRAB isolates by suppressing the expression of $bla_{oxa\text{-}51}$ and

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Table 1. Bacterial strains used in the study.

Strain	Characteristic(s)	Growth condition	Reference/source
Acinetobacter baumannii			
ATCC 17978	Standard strain	Lysogeny broth (LB), 37°C, 180 rpm	American Type Culture Collection
Δ *abaI* ATCC 17978	Autoinducer synthase (abaI) mutant of ATCC 17978	LB, 37°C, 180 rpm	Castañeda-Tamez et al. (2018)
Ci-1 to 3	Clinical isolates from endoscopic tracheal secretions	LB, 37°C, 180 rpm	Maintained in lab
Agrobacterium tumefaciens A136	Biosensor strain for detection of broad range of AHLs	LB + tetracycline (4.5 μg ml^{-1}) + spectinomycin (50 μg ml^{-1}, 30°C, 180 rpm	Fuqua and Winans (1996)

bla$_{oxa\text{-}23}$ β-lactamases, and outer membrane protein *carO* involved in influx of imipenem.

Material and methods

Bacterial strains

Bacterial strains used in this study are described in Table 1. The antimicrobial susceptibility profile of *A. baumannii* strains (Table S1, Supporting Information), was done by the Kirby–Bauer disc diffusion method according to CLSI guidelines (CLSI 2018) using antibiotic discs for different classes of antibiotics (Penicillins, β-lactams, cephalosporins, carbapenems, aminoglycosides, fluoroquinolones, sulphonamide, and polymixins).

Effect of 3IAN on *A. baumannii* growth

The stock solution of 3IAN (1 M) was prepared in dimethyl sulfoxide (DMSO) and stored at −20°C. The effect of different concentrations (0.5–2.5 mM) of 3IAN on the growth of *A. baumannii* ATCC 17978 was determined by growing the cells in the absence and presence of different concentrations of 3IAN at 37°C, 180 rpm for 24 h. The absorbance was measured at 600 nm using the BioTek microplate reader.

Surface motility

For motility assays, LB–LS (medium low salts constituted by 10 g/l tryptone, 5 g/l yeast extract, and 5 g/l NaCl) agar (0.3%) plates with and without 3IAN were prepared. *A. baumannii* strains were inoculated in LB and incubated overnight at 37°C, 180 rpm for 16 h. A volume of 1 μl of overnight cultures were spotted in the center of agar plates, and incubated overnight at 37°C. The diameter of the zone of surface motility was measured (López et al. 2017).

Biofilm formation

Biofilm was formed in 200 μl of LB–LS in a polypropylene microcentrifuge tube under the static condition at 37°C for 24 h in the absence and presence of 3IAN. Before draining the media, planktonic growth was monitored at 600 nm. The biofilm was washed three times with phosphate buffered saline (PBS, pH 7.4) to remove loosely attached bacterial cells. Biofilm was fixed with 200 μl of methanol for 15 min. After removing the methanol, the tubes were air-dried, and stained for 30 min with 200 μl of 0.5% crystal violet, followed by three PBS washings to remove excess stain. The bound stain was extracted with 200 μl of 33% glacial acetic and quantified at 570 nm. OD_{570nm}/OD_{600nm} was used to calculate the change in the biofilm index (BI) (Bhargava et al. 2015).

Confocal laser scan microscopy of biofilm

Biofilm was formed on a glass coverslip placed in flat-bottom petriplate by incubating *A. baumannii* ATCC 17978 in the absence and presence of 3IAN (1 mM) at 37°C for 24 h. After removal of supernatant, the coverslips were rinsed twice with PBS (pH 7.4) and stained with 5 μM Syto™ 9 for 30 min in the dark (Amador et al. 2021). Biofilm was washed three times with PBS, air-dried, and observed under the Nikon Eclipse—NI-E microscope. Syto™ 9 (green) signals were recorded at 517 nm after excitation at 488 nm. COMSTAT function of ImageJ software was used to quantify biomass and surface area (Gautam et al. 2021).

Extracellular polymeric substance production and FTIR spectropscopy

Overnight culture (1%) was inoculated into 100 ml of LB–LS medium with and without 3IAN (1 mM) and incubated for 24 h at 37°C, 180 rpm. Cells were collected by centrifugation at 7000 rpm for 15 min at 4°C after incubation, and cell-free supernatants (CFSs) were kept at −20°C for cell-free extracellular polymeric substance (EPS) extraction. The cell pellets were washed with wash buffer (10 mM Tris/HCl pH 8.0, 10 mM EDTA), resuspended in 100 ml of isotonic extraction buffer (10 mM Tris/HCl pH 8.0, 10 mM EDTA, 2.5 mM NaCl), and incubated at 4°C for 12 h. Supernatant containing cell-bound EPS was collected by centrifugation at 5000 rpm for 15 min at 4°C. To precipitate EPS, cell-free culture supernatants containing cell-free EPS and isotonic buffer containing cell-bound EPS were combined, mixed with three volumes of ice-cold ethanol, and stored at −20°C for 18 h. Precipitated EPS was collected by centrifugation at 10,000 rpm for 10 min at 4°C, vacuum dried, and analyzed by FTIR spectrometry (Spectrum RX-IFTIR, PerkinElmer) at 650–4000 cm^{-1} (Sethupathy et al. 2020).

Acyl homoserine lactone extraction and quantitation

Acyl homoserine lactone (AHL) was extracted from the CFS of *A. baumannii* ATCC 17978 culture grown in LB–LS medium in the absence or presence of 3IAN (1 mM) for 24 h under static conditions. The CFS was extracted twice with equal volumes of acidified ethyl acetate (glacial acetic acid, 0.01% v/v) and stored at 4°C for overnight. The upper organic layer containing extracted AHLs was separated, dried at 40°C in a rotary evaporator (Ika, Cole Parmer), and resuspended in DMSO (Gautam et al. 2021).

For plate bioassay, 10 ml of soft agar (0.8%) containing spectinomycin (50 μg ml^{-1}), tetracycline (4.5 μg ml^{-1}), and X-gal (50 μg ml^{-1}) was seeded with stationary phase culture of *A. tumefaciens* A136 (OD_{600}~1) and overlaid on a thin layer of hard agar (2%). Wells were bored in the top layer of soft agar and 25 μl of extracted AHLs was added. The plate was incubated at 30°C for 124 h

Table 2. Primers used in the study.

Gene	Forward primer	Reverse primer
16S rRNA	GAGGAAGGTGGGGATGACGT	AGGCCCGGGAACGTATTCAC
abaI	CAGCTTATGTCGTGGCTCAAG	TGATGACACAGCCTGACTGC
adeB	GGATTATGGCGACTGAAGGA	AATACTGCCGCCAATACCAG
adeG	GCGTTGCTGTGACAGATGTT	TTGTGCACGGACCTGATAAA
adeJ	GCGAATGGACGTATGGTTCT	CATTGCTTTCATGGCATCAC
abeS	ATACGGTTTGGACAGGCATTG	CGGCAATCATTCTCATCGCA
abeM	TGCCAATTGGTTTAGCTGTG	TACTTGGTGTGCGGCAATAA
aceI	CACCCTCAAAACCAATCGCA	TCGGTGTGTTTATGGCAGTG
A1S_1503	GCCAAGAAAGTAAGTCGCGT	ACCAAAATGCGGCAACAAGA
msc	AGTGGTGGGCTTTATTGGCT	ACCCACTTCGCTGCTTAACT
emrB	AGGAATGAGCCACGCAAGAA	TGGGCAAGTCCGTGAAATGA
emrA	CTGTGCCTGAGCCGTTAAAT	AAGTCTGCACGGTCGTCAAT
carO	AAAGCACCACCGTAACCTGT	GTGCTGCAATGGCTGATGAA
blaoxa-23	GATCGGATTGGAGAACCAGA	ATTTCTGACCGCATTTCCAT
blaoxa-51	CAAATCACAGCGCTTCAAAA	TTTTGAAGGTCGAAGCAGGT

and observed for the appearance of a blue colour around the well (Bhargava et al. 2015).

AHL was quantified by liquid bioassay using a standard curve prepared with pure 3-OH-C12-HSL. *A. tumefaciens* A136 was grown in the presence of 3-OH-C12-HSL ranging from 40 to 160 μM and extracted AHLs at 30°C, 180 rpm for 16 h. To 200 μl of each culture, 50 μl of 0.1% SDS and 100 μl chloroform was added, vortexed for 30 s, and immediately incubated on ice for 20 min. The suspension was centrifuged at 12,000 rpm for 5 min at 4°C. A volume of 100 μl of clear supernatant was added to 50 μl of ONPG (4 mg ml^{-1}) and incubated at 30°C for 30 min. The reaction was stopped by adding 80 μl of 1 M Na_2CO_3 and the OD was measured at 420 and 550 nm (Lau et al. 2020). Miller units were determined using the formula:

$$\text{Miller unit} \quad (\text{MU}) = 1000 \times \frac{\text{OD420} - (1.75 \times \text{OD550})}{\text{OD600} \times \text{Vol} \ (\text{ml}) \times \text{Time} \ (\text{min})}.$$

Determination of minimum inhibitory concentration

The minimum inhibitory concentration (MIC) of antibiotics (ofloxacin, imipenem, ciprofloxacin, tobramycin, and levofloxacin) against *A. baumannii* was determined using the broth microdilution method as per CLSI guidelines (CLSI 2018). A single colony of the bacterial strain was inoculated into Mueller Hinton (MH) broth medium and incubated at 37°C, 180 rpm for 16 h. Overnight grown bacterial culture was diluted in fresh MH broth to a turbidity equivalent to 0.5 McFarland standard, mixed with different concentrations of antibiotics, and incubated for 24 h at 37°C, 180 rpm. After incubation, the cell density of the culture was measured at 600 nm using the BioTek multimode plate reader. The concentration at which no visible growth was observed as taken as MIC.

A. baumannii was grown in MH broth in the absence and presence of varying concentrations of antibiotics at 37°C for 16 h. The ability of 3IAN to reduce the MIC of antibiotics was calculated as the fold-reduction in MIC and represented as the modulation factor (MF) defined as $MIC_{Drug}/MIC_{Drug\ in\ combination}$ (Christena et al. 2015).

Ethidium bromide accumulation assay

A. baumannii ATCC 17978 culture grown for 6 h was harvested and resuspended in PBS (pH 7.4) containing 0.4% glucose. The cells (200 μl) were transferred to a Genetix 96-well black microtiter plate, before adding 3IAN (1 mM) or CCCP (100 μM). Ethidium bromide (EtBr; 2 μg ml^{-1}) was then added, and its fluorescence was measured using a BioTek multimode plate reader with an excitation wavelength of 544 nm and emission wavelength of 590 nm (Kern et al. 2006). Fluorescence was normalized using the growth (OD_{600}) of the individual samples.

EtBr efflux assay

A. baumannii ATCC 17978 culture grown for 6 h was harvested, washed, and resuspended in PBS before being treated with 1 mM 3IAN or 100 μM CCCP (positive control) along with EtBr (2 μg ml^{-1}) and glucose (0.4%). The untreated cell suspension was processed in the similar way and used as the control. For maximal uptake of EtBr, the samples were incubated at 37°C for 1 h. The cells were harvested, and resuspended in PBS after being washed twice to eliminate any remaining EtBr. The EtBr efflux was calculated by measuring its fluorescence using a BioTek multimode plate reader with an excitation wavelength of 544 nm and emission wavelength of 590 nm (Christena et al. 2015).

Quantitative real time PCR

A 1% overnight culture of *A. baumannii* was inoculated into 10 ml LB–LS medium and then cultured at 37°C, 180 rpm for 6 h in the absence and presence of 3IAN (1 mM). Total RNA was isolated using TRI reagent (Sigma Aldrich) as per the manufacturer's instructions. RNA (1 μg) was reverse transcribed using Verso cDNA synthesis kit (Thermo Scientific). The expression of autoinducer synthase (*abaI*) gene, efflux pump genes (*adeB, adeG, adeJ, abeS, abeM, aceI, A1S_1503, msc, emrB*, and *emrA*), gene of outer membrane protein *carO,* and genes encoding for β-lactamases ($bla_{oxa\text{-}51}$ and $bla_{oxa\text{-}23}$) was determined using iTaq Universal SYBR Green supermix (BioRad) in CFX96 Touch Deep Well Real-Time PCR Detection System (Bio-Rad). The primers used for qRT-PCR are listed in Table 2. 16S rRNA was used as an internal control. The relative gene expression was calculated using $2^{(-\Delta\Delta Ct)}$ method (Pfaffl 2006).

Cytotoxicity of 3IAN

The effect of 3IAN on the viability of HeLa cells was determined using the MTT assay (Sidhu and Capalash 2021). HeLa cells (1×10^4 cells well^{-1}) seeded in a 96-well plate were incubated for overnight and treated with two different concentrations (0.5 and 1 mM) of 3IAN for 24 h. Subsequently, MTT (0.5 mg ml^{-1}) solution was added and the cells were incubated for 3 h. The

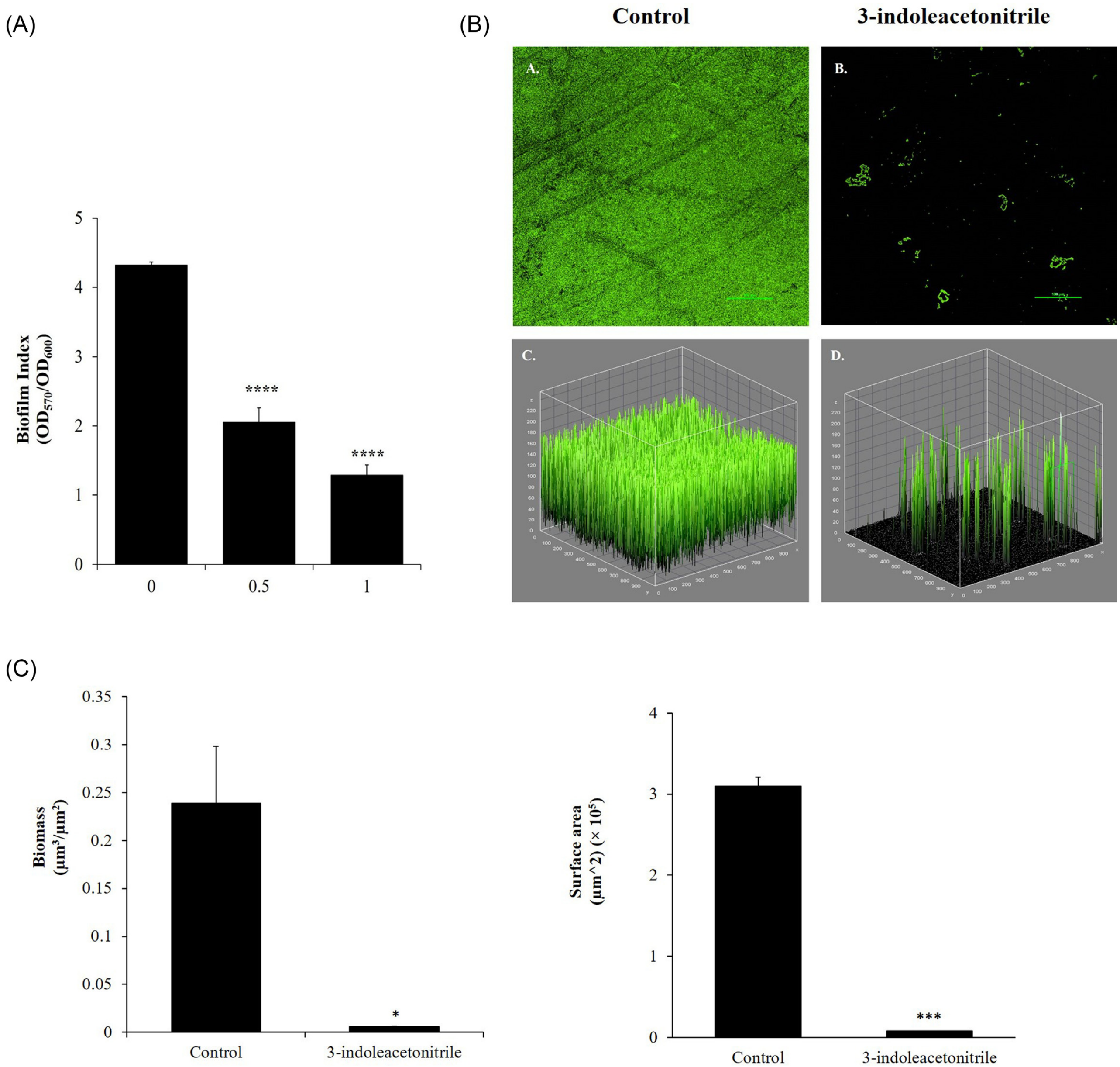

Figure 1. (A) Effect of 3IAN on biofilm of *A. baumannii* ATCC 17978. Data are represented as mean ± SD of three independent experiments. (B) Confocal Laser Scanning Microscopy images of *A. baumannii* ATCC 17978 biofilm formed on glass coverslip in the absence and presence of 1 mM 3IAN stained by Syto9. Top panel images are 2D images while the lower panel images are isosurface-rendered 3D surface plots of the same data set, using ImageJ software (version 1.53q). Images were taken at 20× magnification. (C) Quantification of biomass and surface area of *A. baumannii* ATCC 17978 biofilm formed in the presence of 3IAN, using COMSTAT (version 2.1). Data are represented as mean ± SD of two independent experiments.*$P \leq 0.05$; ***$P \leq 0.001$.

medium was decanted and formazan crystals were dissolved in 200 µl DMSO. The absorbance at 570 nm was measured using a microplate reader (BioTek) and the percentage of cell survival was evaluated using the formula:

$$\text{Percent cell survival} = \frac{Absorbance\,(Test) - Absorbance\,(Blank)}{Absorbance\,(Control) - Absorbance\,(Blank)} \times 100.$$

Statistical analysis

GraphPad Prism 5.01 (GraphPad Sofware Inc., CA, USA) was used to perform statistical analysis on the data. Student's *t*-test and one-way ANOVA were used to analyze the data. *P*-value of ≤ 0.05 was considered statistically significant. The data has been expressed as mean ± standard deviation (SD) with at least three independent experiments.

Results

3IAN inhibited biofilm formation and surface motility in *A. baumannii*

3IAN showed a growth inhibitory effect on *A. baumannii* ATCC 17978 in a concentration-dependent manner. No growth was visible at 2.5 mM of 3IAN while at 0.5 and 1 mM, 89.5% and 80.6%

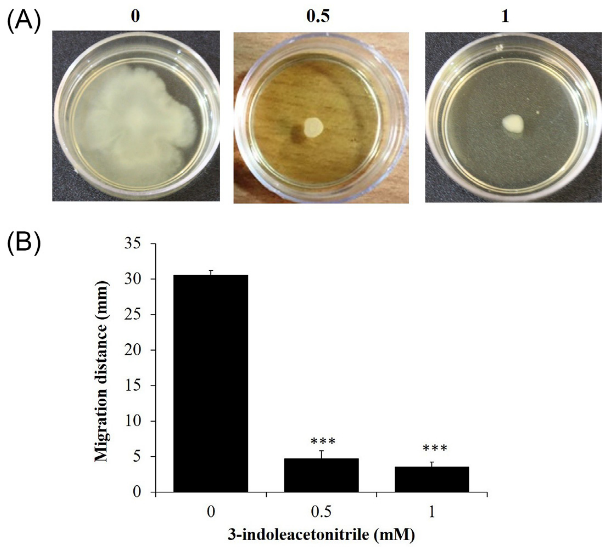

Figure 2. (A) Surface motility of *A. baumannii* ATCC 17978 in the absence and presence of 3IAN (0.5 and 1 mM), and (B) the bar graph represents surface motility diameter (mm). Data are represented as mean ± SD of two independent experiments. ***$P \leq 0.001$.

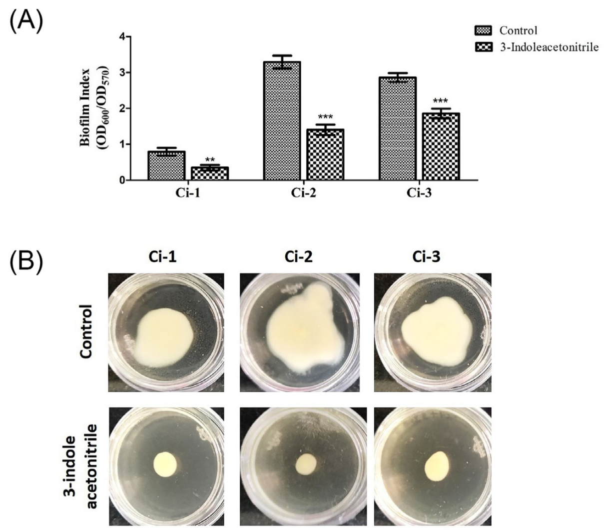

Figure 3. Effect of 3IAN on (A) biofilm formation, and (B) surface motility of CRAB isolates. Data are represented as mean ± SD of three independent experiments. **$P \leq 0.01$; ***$P \leq 0.001$

survival was observed, respectively, as compared to the control (Figure S1, Supporting Information).

3IAN exhibited antibiofilm activity against *A. baumannii* ATCC 17978. At subinhibitory concentrations of 0.5 and 1 mM, it reduced biofilm formation by 49% and 71%, respectively, compared to the untreated cells (Fig. 1A). Confocal Laser Scan Microscopy (CLSM) also revealed significant reduction in biofilm in the presence of 1 mM 3IAN (Fig. 1B). COMSTAT analysis showed that biofilm formed in the presence of 3IAN had significantly reduced the biomass and surface area by 33-fold and 39-fold, respectively, compared to the untreated biofilm (Fig. 1C). The EPS, which is an important component of biofilm was also reduced. The FTIR spectrum of EPS in 3IAN-treated culture showed reduction in absorption intensities of polysaccharides (1200–900 cm^{-1}), proteins (1500–1700 cm^{-1}), and lipids (2850–3020 cm^{-1}) in comparison to untreated cells (Figure S2, Supporting Information). Motility, which plays an important role in biofilm formation in *A. baumannii*, was also significantly reduced by 83.5% and 88% in the presence of 3IAN at 0.5 mM and 1 mM, respectively (Fig. 2). A total of three CRAB clinical isolates (Ci-1, Ci-2, and Ci-3) also showed significant inhibition of biofilm formation by 55.8%, 57.3%, and 35%, respectively in the presence of 1 mM 3IAN (Fig. 3A). Similarly, there was significant reduction in their surface motility by > 80% compared to the untreated cultures (Fig. 3B).

Inhibition of QS by 3IAN

Since biofilm is controlled by the QS system in *A. baumannii*, the effect of 3IAN on the expression of the autoinducer synthase gene *abaI* and AHL production was determined. 3IAN significantly reduced the expression of the *abaI* gene by 3.36-fold (Fig. 4A) and the production of AHL by 3.74-fold (Fig. 4B; Figure S3, Supporting Information). Further to confirm the role of QS in biofilm inhibition by 3IAN, *A. baumannii* ATCC 17978 *abaI* deletion mutant was used. In comparison to the wildtype strain, the *abaI* mutant displayed 90% biofilm inhibition and exogenous addition of 3-OH-C12-HSL (5 μM) restored biofilm formation by 27%. 3IAN at subinhibitory concentration had no effect on the biofilm of *abaI* mutant, but it reduced the restoration of biofilm formation when 3-OH-C12-HSL was added exogenously (Fig. 4C).

3IAN decreased MIC of antibiotics against *A. baumannii* ATCC 17978

In order to determine the use of 3IAN as adjuvant for antibiotics, its effect on the MIC of different classes of antibiotics against *A. baumannii* ATCC 17978 was determined. The presence of 3IAN at subinhibitory concentration (1 mM) effectively reduced the MIC of antibiotics. The MIC of levofloxacin was reduced from 0.125 to 0.06 μg ml^{-1} with an MF of 2, tobramycin from 2 to 0.5 μg ml^{-1} with an MF of 4, ciprofloxacin from 0.25 to 0.063 μg ml^{-1} with an MF of 4, and ofloxacin from 1 to 0.125 μg ml^{-1} with an MF of 8. The MIC of imipenem was also reduced from 0.25 to 0.031 μg ml^{-1} with an MF of 8 in the presence of 3IAN.

3IAN inhibited efflux pumps in *A. baumannii* ATCC 17978

To study the involvement of efflux pumps in increasing susceptibility to antibiotics in *A. baumannii* ATCC 17978 by 3IAN, EtBr efflux and accumulation assays were performed. 3IAN was found to significantly ($P < 0.0001$) inhibit the efflux of EtBr as shown by 2-fold increased fluorescence, which was comparable to that observed with the efflux pump inhibitor carbonyl cyanide 3-chlorophenylhydrazone (CCCP; 2.01-fold) after 60 s of exposure (Fig. 5A). Likewise, 3IAN significantly ($P < 0.0001$) enhanced EtBr accumulation (2.16-fold) in *A. baumannii* ATCC 17978 cells after 60 s of exposure, equivalent to CCCP-treated cells (2.91-fold; Fig. 5B).

3IAN led to significant downregulation in the expression of *adeB, adeG,* and *adeJ* genes encoding for RND pumps by 1.94-fold, 5.14-fold, and 3.59-fold, respectively. Expression of SMR family gene *abeS* and MATE family gene *abeM* was also downregulated by 3.25-fold and 3.37-fold, respectively. Genes encoding for PACE pumps *aceI,* and *A1S_1503* were significantly downregulated by 2.5-fold and 2.69-fold, respectively. Maximum downregulation of 9.52-fold was observed in the expression of *msc* gene encoding for mechanosensitive (MS) channel. Expression of *emrB* was downreg-

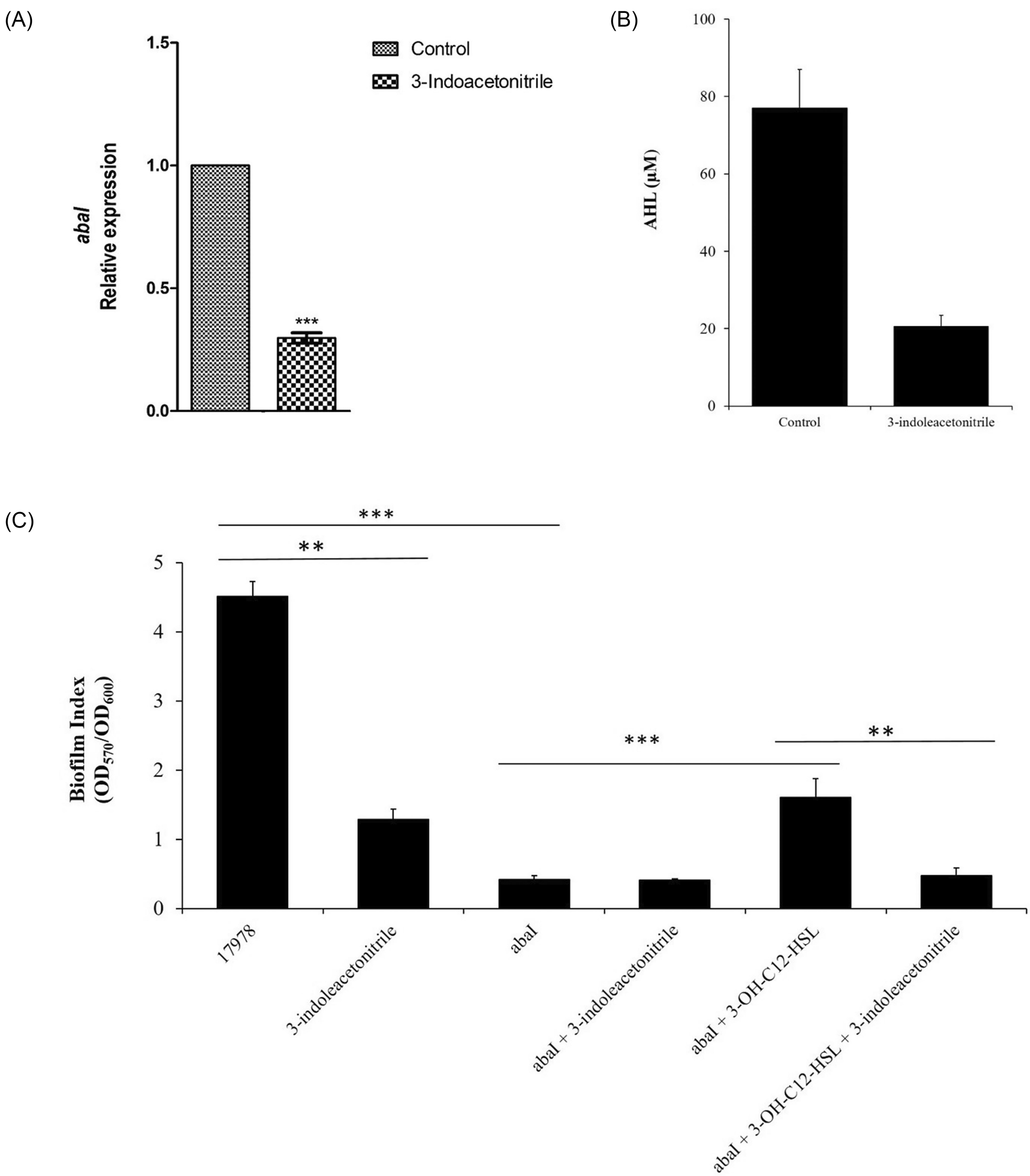

Figure 4. (A) Relative expression of autoinducer synthase (*abaI*) gene in *A. baumannii* ATCC 17978 cells grown in the absence and presence of 3IAN (1 mM) for 6 h at 37°C, 180 rpm. 16S rRNA was used to normalize the expression of the gene. Untreated cells were taken as control with basal level expression indicated as 1. Data are representative of three independent experiments. $^{***}P \leq 0.001$. (B) Production of AHL by *A. baumannii* ATCC 17978 in the absence and presence of 3IAN. Data are representative of two independent experiments. $^{*}P \leq 0.05$ (C) Effect of 3IAN on biofilm formation of *A. baumannii* ATCC 17978 and its *abaI* deletion mutant. Data are representative of three independent experiments. $^{**}P \leq 0.01$; $^{***}P \leq 0.001$.

ulated by 4.15-fold, while *emrA* showed insignificant decrease in the expression (Fig. 5C).

3IAN increased sensitivity of CRAB isolates to imipenem

Since the presence of 3IAN reduced MIC of imipenem in *A. baumannii* ATCC 17978, its effect on the MIC of imipenem in carbapenem-resistant isolates was investigated. In the presence of 3IAN (1 mM), the MIC of imipenem decreased by 4-fold, from 32 to 8 μg/ml, in all the three carbapenem-resistant isolates (Fig. 6). Disc diffusion assay also revealed zone of inhibition around the resistant strains (Figure S4, Supporting Information).

3IAN was found to significantly reduce the expression of bla_{OXA-51} (5.9–43.4 fold) and bla_{OXA-23} genes (1.9–2.2 fold) in all three

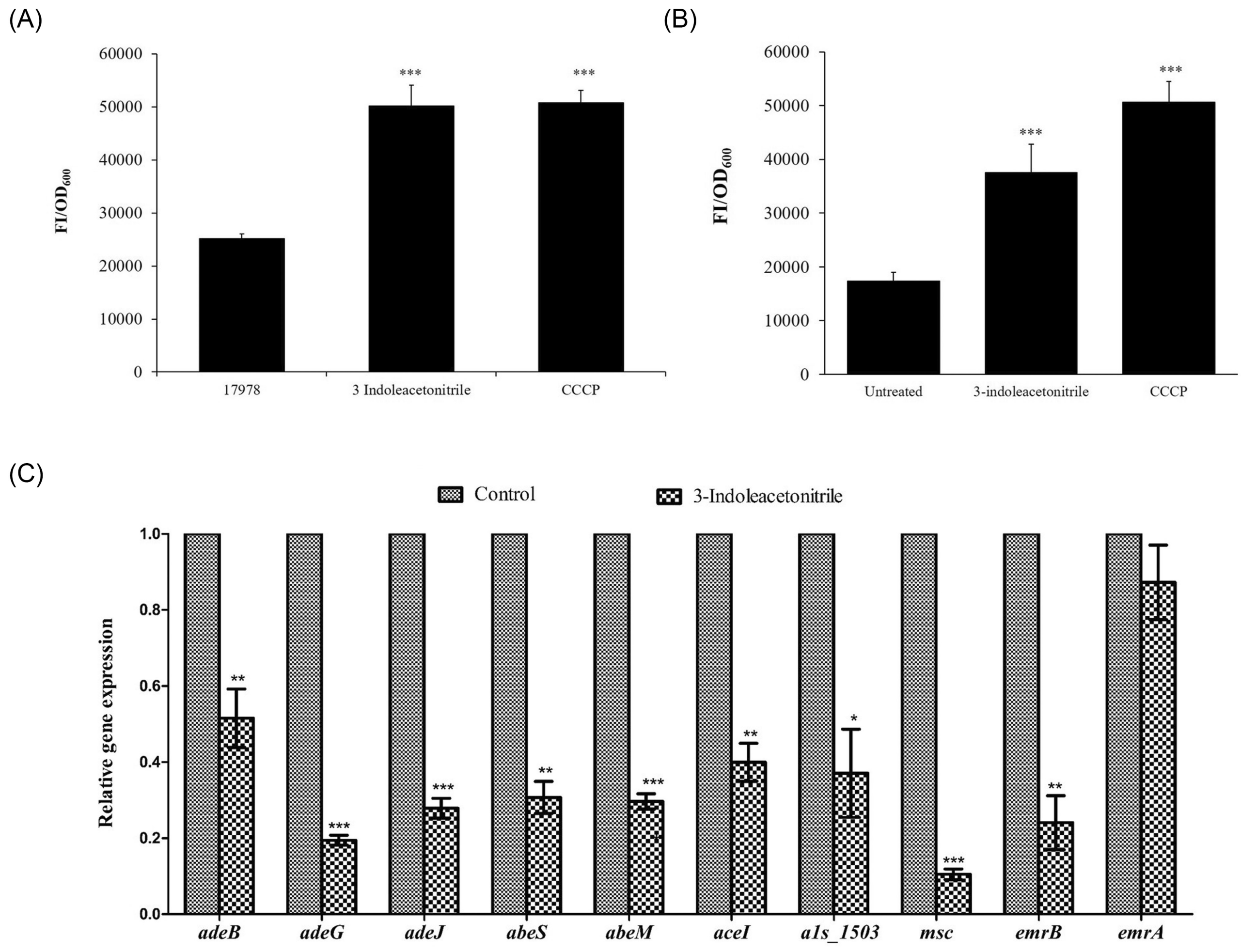

Figure 5. (A) Accumulation of EtBr and (B) efflux of EtBr in *A. baumannii* ATCC 17978 in the presence of 1 mM 3IAN and CCCP (positive control) after 60 s of exposure. (B) Relative expression of efflux pump-encoding genes (*adeB, adeG, adeJ, abeS, abeM, aceI, A1S_1503 msc, emrB,* and *emrA*) in cells grown in the absence and presence of 3IAN at 1 mM for 6 h at 37°C, 180 rpm. 16S rRNA gene was used to normalize the expression of the genes. Untreated cells were taken as control with basal level expression indicated as 1. Data are representative of three independent experiments. Bars represent the mean ± SD. $^*P \leq 0.05$; $^{**}P \leq 0.01$; $^{***}P \leq 0.001$; $^{****}P \leq 0.0001$

carbapenem-resistant isolates. In addition, there was significant downregulation in the expression of *adeB* gene (2.8–18.5 fold) encoding for the efflux pump involved in the extrusion of imipenem and outer membrane protein *carO* gene (1.3–5.68 fold) associated with the influx of imipenem (Fig. 7).

3IAN-reduced persister cells against imipenem in *A. baumannii* ATCC 17978

Concentration-dependent assay of exponential phase *A. baumannii* ATCC 17978 cells with increasing concentrations of imipenem showed biphasic killing curve, with initial rapid killing of the majority of cells levelling off at 20 × MIC (MIC = 0.25 μg ml^{-1}) followed by a plateau of surviving persister cells (Fig. 8A). The time-dependent assay with 20 × MIC of imipenem revealed the presence of 0.32% persister cells at 5 h, which showed rebounding (survivors outgrowth) at 24 h. Treatment of exponential phase cells with a combination of 3IAN and imipenem, on the other hand showed significant (2log-fold) reduction in persister cells with no rebounding growth of persister cells at 24 h (Fig. 8B).

Cytotoxicity of 3IAN on HeLa cell line

The viability of HeLa cells was not affected at 0.5 and 1 mM concentrations of 3IAN used in this study (Figure S5, Supporting Information).

Discussion

The rise of multidrug resistant *A. baumannii* strains has become a global problem reducing the effective treatment options. Antibiotic resistance is often linked to biofilm formation and persistence, leading to infection recurrence (Nguyen and Joshi 2021). Indole-based derivatives have attracted attention as potential drug candidates since they influence QS, biofilm formation, antibiotic tolerance, and virulence factor expression (Lee et al. 2015, Kumar et al. 2021). In the present study, 3IAN, a derivative of indole, decreased surface motility and biofilm formation in *A. baumannii* strains at subinhibitory concentration. 3IAN has been shown to inhibit biofilm formation in *E. coli* O157:H7 and *P. aeruginosa* (Lee et al. 2011). 5-Iodoindole has been shown to be effective in inhibiting biofilm formation in *A. baumannii* strains (Raorane et al. 2020), but the possible mechanism(s) has not been elucidated. QS plays a significant role in biofilm formation in *A. baumannii* (Saipriya

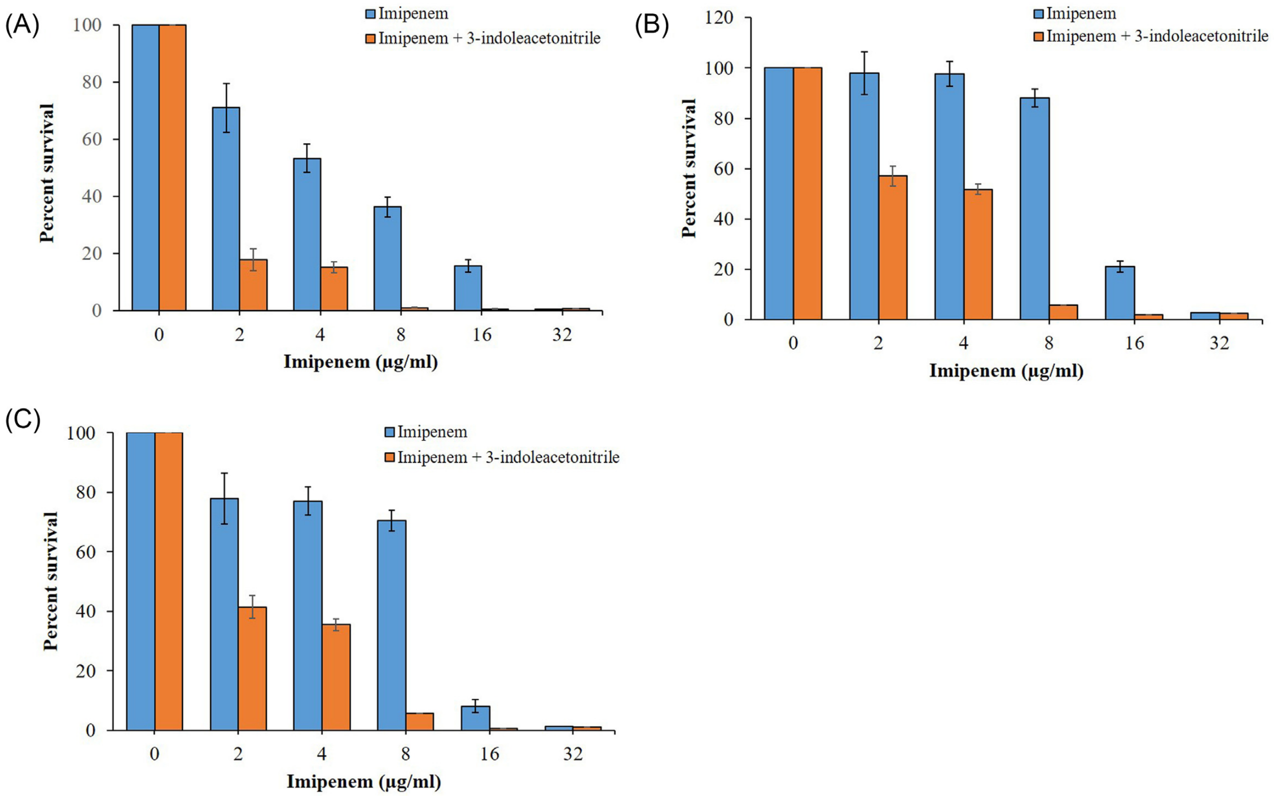

Figure 6. Effect of 3IAN (1 mM) on the susceptibility of (A) Ci-1, (B) Ci-2, and (C) Ci-3 isolates to imipenem. Data are represented as mean ± SD of two independent experiments.

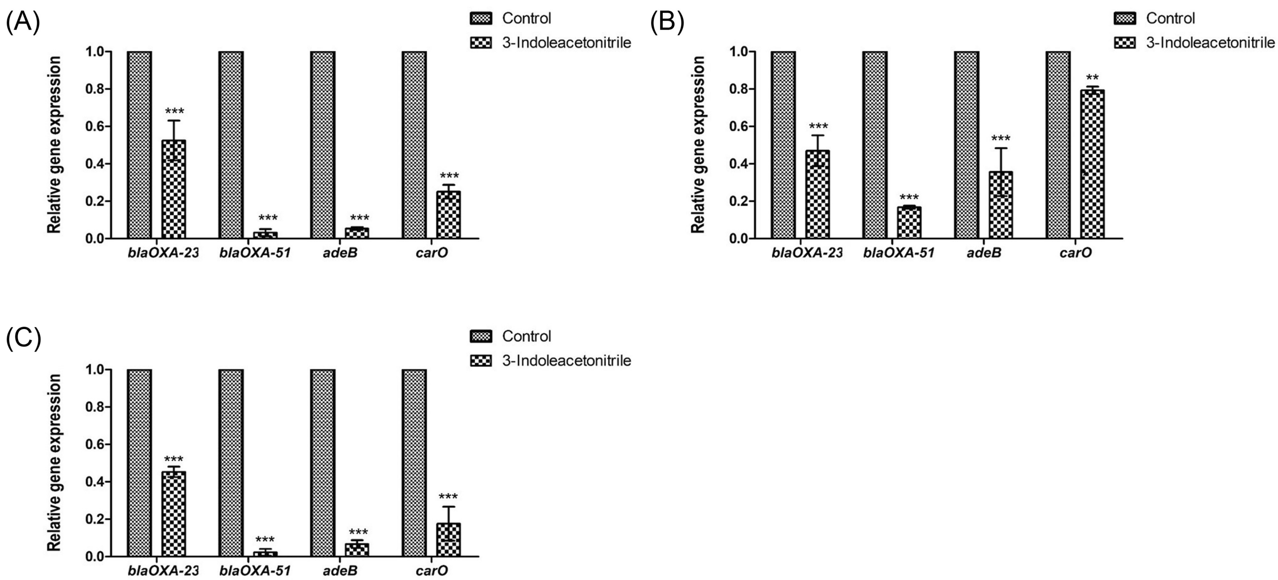

Figure 7. Relative expression of *bla*$_{OXA-51}$, *bla*$_{OXA-23}$, *adeB*, and *carO* genes in (A) Ci-1, (B) Ci-2, and (C) Ci-3 isolates grown in the absence and presence of 3IAN at 1 mM for 6 h at 37°C, 180 rpm. 16S rRNA gene was used to normalize the expression of the genes. Untreated cells were taken as control with basal level expression indicated as 1. Data are representative of two independent experiments. Bars represent the mean ± SD. $^{**}P \leq 0.01$; $^{***}P \leq 0.001$

et al. 2020). Autoinducer synthase (*abaI*) produces QS signal 3-OH-C_{12}-HSL in *Acinetobacter* spp., required for biofilm formation (He et al. 2015). 3IAN inhibited QS in *A. baumannii* by reducing the expression of autoinducer synthase *abaI,* resulting in a significant decrease in AHL production, which might be responsible for reduction in biofilm formation. Insignificant effect on biofilm formation in *abaI* mutant by 3IAN, but reduced reversion of biofilm formation on addition of exogenous 3-hydroxy-dodecanoyl-homoserine lactone (3-OH-C_{12}-HSL) supported QS as the target of 3IAN.

Combinatorial drug therapy is an emerging strategy to prevent bacterial infections as it offers the advantage of decreasing

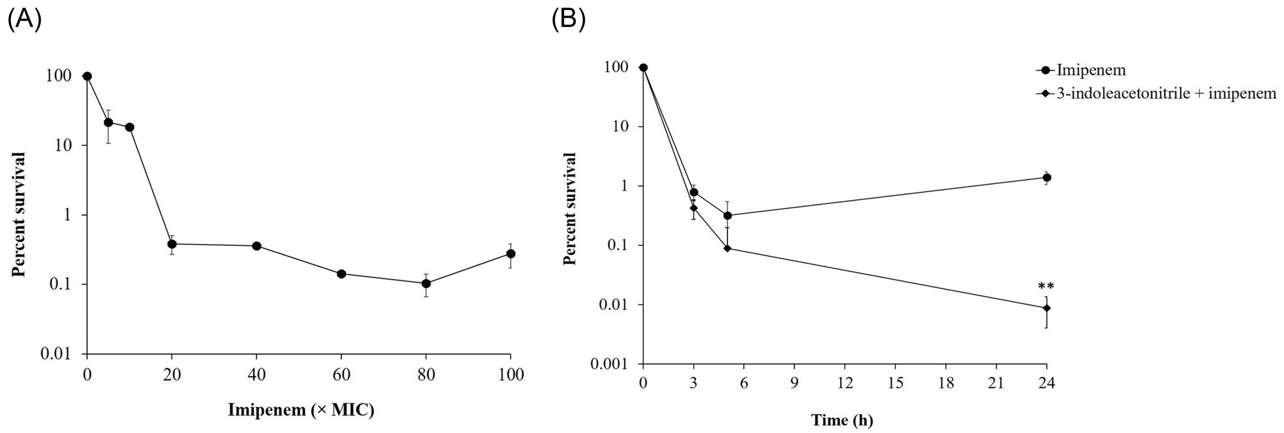

Figure 8. (A) Exponential phase cells of *A. baumannii* ATCC 17978 on treatment with different concentrations of imipenem for 3 h. (B) Effect of 3IAN on persister cells formation of exponential phase cells against 20 × imipenem for different time intervals. Untreated cells at 0 h ($2.56 \pm 0.15 \times 10^8$ CFU/ml) were taken as 100% and % survival was calculated accordingly. Data are represented as mean log % survival ± SD of three independent experiments. $^{**}P \leq 0.01$

drug dosage (Angst et al. 2021). Previous studies have shown that antibiofilm agents increased the sensitivity of bacteria to antibiotics (Mishra et al. 2020, Selvaraj et al. 2020). In the present study, 3IAN reduced the MIC of imipenem, ofloxacin, ciprofloxacin, tobramycin, and levofloxacin against *A. baumannii* ATCC 17978, indicating that it is a broad-acting antibiotic sensitizing compound. This ability of 3IAN to enhance the antibacterial activity of commonly used antibiotics should be of high significance in the clinical context where infection due to MDR *A. baumannii* is on the rise. The increased antibiotic sensitivity in the presence of 3IAN is probably mediated through inhibition of efflux pumps in *A. baumannii*. Over expression of efflux pumps is one of the main reasons for multidrug resistance to various antimicrobial agents in *A. baumannii* (Abdi et al. 2020). AdeABC, AdeFGH, and AdeIJK efflux pumps of resistance nodulation-division (RND) family, AbeS efflux pump belonging to small multidrug resistance family (SMR) and AbeM efflux pump of multidrug and toxic compound extrusion family (MATE) are associated with fluoroquinolone resistance in *A. baumannii* (Kornelsen and Kumar 2021, Roy et al. 2021). A significant decrease in *adeB, adeG, adeJ, abeS*, and *abeM* expression by 3IAN might have resulted in increased sensitivity to ofloxacin, ciprofloxacin, and levofloxacin against *A. baumannii* ATCC 17978. The efflux of aminoglycosides and imipenem in *A. baumannii* has been linked to the AdeABC efflux pump (Magnet et al. 2001, Amiri et al. 2019). Proteobacterial antimicrobial compound efflux (PACE) pumps has been shown to be responsible for the efflux of chlorhexidine and acriflavine, while the EmrAB efflux pump contributed to colistin resistance in *A. baumannii* (Hassan et al. 2015, Lin et al. 2017). Since expression of these genes was significantly downregulated by 3IAN it might affect the sensitivity of *A. baumannii* to disinfectants and colistin. Mechanosensitive (MS) ion channels are integral membrane proteins (Ridone et al. 2019), known to mediate hypoionic shock-induced aminoglycoside potentiation in *E. coli* (Lv et al. 2021). Since their function in *A. baumannii* is unknown, further studies are required to understand their role in antibiotic resistance in *A. baumannii*.

Carbapenem resistance has increased in clinical *A. baumannii* isolates worldwide (Şimşek and Demir 2020), warranting the discovery of novel compounds to control infections. Interestingly, 3IAN reduced the MIC of imipenem in CRAB isolates. OXA-51-like and OXA-23-like carbapenemases have been associated with imipenem resistance in *A. baumannii*, where insertion sequence IS*Aba1* plays a major role in their transfer and expression (Vijayakumar et al. 2020). Reduction in the expression of both $bla_{\text{OXA-51}}$ and $bla_{\text{OXA-23}}$ genes by 3IAN with respect to IS*Aba1* requires further investigation. Moreover, 3IAN showed multiple effects as it reduced the expression of transporter protein *adeB* and outer membrane protein *carO,* which have a vital role in the efflux and influx of imipenem respectively, in *A. baumannii* (Zhu et al. 2019, 2020).

Biofilms provide favourable environment for persister cells (Hossain et al. 2021). Persister cells within biofilm are major contributors to treatment failures and recurrent infections (Wood 2016). *A. baumannii* has been shown to form persister cells in response to different classes of antibiotics (Kaur et al. 2018, Kashyap et al. 2021). We also observed persister cells formation in *A. baumannii* against imipenem, consistent with a previous study (Chung and Ko 2019). However, combination of 3IAN with imipenem effectively reduced persister cells formation and prevented persisters from rebounding (survivors outgrowth). Bacterial persisters use a two-pronged strategy to sustain antibiotic attack by slowing down most physiological functions and simultaneously activating their efflux systems to remove intracellular antibiotics, resulting in tolerance (Pu et al. 2017). Curcumin has been shown act synergistically with colistin by decreasing efflux, resulting in increased efficacy and decreased persister formation in *A. baumannii* (Kaur et al. 2018). A significant downregulation of the efflux pump genes by 3IAN might have contributed to the increased lethality and low persistence.

Antibiofilm activity of 3IAN at subinhibitory concentration with less chances of drug resistance development, reduced persistence, and resistance to imipenem, and no cytotoxic effect on HeLa cells warrants *in vivo* studies to evaluate the therapeutic efficacy of 3IAN in combination with conventional antibiotics against *A. baumannii*.

Supplementary data

Supplementary data are available at *FEMSPD* online.

Authors' contributions

S.K.: conceptualization, data curation, formal analysis, investigation, methodology, validation, visualization, writing—original draft, writing—review, and editing. H.S.: .methodology, formal analysis, and writing of cytotoxicity assay. P.S.: conceptualization, writing—review, and editing. N.C.: conceptualization, methodology, funding acquisition, resources, project administration, supervision, writing—review, and editing. All authors have read and approved the final manuscript.

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References

Abdi SN, Ghotaslou R, Ganbarov K *et al.* *Acinetobacter baumannii* efflux pumps and antibiotic resistance. *Infect Drug Resist* 2020;**13**:423.

Amador CI, Stannius RO, Røder HL *et al.* High-throughput screening alternative to crystal violet biofilm assay combining fluorescence quantification and imaging. *J Microbiol Methods* 2021;**190**:106343.

Amiri G, Shaye MA, Bahreini M *et al.* Determination of imipenem efflux-mediated resistance in *Acinetobacter* spp., using an efflux pump inhibitor. *Iran J Microbiol* 2019;**11**:368.

Angst DC, Tepekule B, Sun L *et al.* Comparing treatment strategies to reduce antibiotic resistance in an in vitro epidemiological setting. *Proc Natl Acad Sci* 2021;**118**:e202346711.

Ayobami O, Willrich N, Harder T *et al.* The incidence and prevalence of hospital-acquired (carbapenem-resistant) *Acinetobacter baumannii* in Europe, Eastern Mediterranean and Africa: a systematic review and meta-analysis. *Emergi Microbes Infect* 2019;**8**:1747–59.

Ayoub Moubareck C, Hammoudi Halat D. Insights into *Acinetobacter baumannii*: a review of microbiological, virulence, and resistance traits in a threatening nosocomial pathogen. *Antibiotics* 2020;**9**:119.

Bhargava N, Singh SP, Sharma A *et al.* Attenuation of quorum sensing-mediated virulence of *Acinetobacter baumannii* by Glycyrrhiza glabra flavonoids. *Fut Microbiol* 2015;**10**:1953–68.

Carvalho G, Balestrino D, Forestier C *et al.* How do environment-dependent switching rates between susceptible and persister cells affect the dynamics of biofilms faced with antibiotics?. *Npj Biofilms Microbiomes* 2018;**4**:1–8.

Castañeda-Tamez P, Ramírez-Peris J, Pérez-Velázquez J *et al.* Pyocyanin restricts social cheating in *Pseudomonas aeruginosa*. *Front Microbiol* 2018;**9**:1348.

Christena LR, Subramaniam S, Vidhyalakshmi M *et al.* Dual role of pinostrobin-a flavonoid nutraceutical as an efflux pump inhibitor and antibiofilm agent to mitigate food borne pathogens. *RSC Adv* 2015;**5**:61881–7.

Chung ES, Ko KS. Eradication of persister cells of *Acinetobacter baumannii* through combination of colistin and amikacin antibiotics. *J Antimicrob Chemother* 2019;**74**:1277–83.

CLSI. 2018 *Performance Standards for Antimicrobial Susceptibility Testing*. 28th ed. CLSI supplement M100. Wayne.

de Vos M, Kriksunov KL, Jander G. Indole-3-acetonitrile production from indole glucosinolates deters oviposition by *Pieris rapae*. *Plant Physiol* 2008;**146**:916–26.

Fuqua C, Winans SC. Conserved cis-acting promoter elements are required for density-dependent transcription of *Agrobacterium tumefaciens* conjugal transfer genes. *J Bacteriol* 1996;**178**:435–40.

Gautam LK, Sharma P, Capalash N. Attenuation of *Acinetobacter baumannii* virulence by inhibition of polyphosphate kinase 1 with repurposed drugs. *Microbiol Res* 2021;**242:** 126627.

Hassan KA, Liu Q, Henderson PJ *et al.* Homologs of the *Acinetobacter baumannii AceI* transporter represent a new family of bacterial multidrug efflux systems. *MBio* 2015;**6**:e01982–14.

He X, Lu F, Yuan F *et al.* Biofilm formation caused by clinical *Acinetobacter baumannii* isolates is associated with overexpression of the AdeFGH efflux pump. *Antimicrob Agents Chemother* 2015;**59**:4817–25.

Hossain T, Deter HS, Peters EJ Antibiotic tolerance, persistence, and resistance of the evolved minimal cell, *Mycoplasma mycoides* JCVI-Syn3B. *Iscience* 2021;**24**:102391.

Jain M, Sharma A, Sen MK *et al.* Phenotypic and molecular characterization of *Acinetobacter baumannii* isolates causing lower respiratory infections among ICU patients. *Microb Pathog* 2019;**128**: 75–81.

Kashyap S, Kaur S, Sharma P *et al.* Combination of colistin and tobramycin inhibits persistence of *Acinetobacter baumannii* by membrane hyperpolarization and down-regulation of efflux pumps. *Microbes Infect* 2021;**23**:104795.

Kaur A, Sharma P, Capalash N. Curcumin alleviates persistence of *Acinetobacter baumannii* against colistin. *Sci Rep* 2018;**8**:1–1.

Kern WV, Steinke P, Schumacher A *et al.* Effect of 1-(1-naphthylmethyl)-piperazine, a novel putative efflux pump inhibitor, on antimicrobial drug susceptibility in clinical isolates of *Escherichia coli*. *J Antimicrob Chemother* 2006;**57**:339–43.

Kornelsen V, Kumar A. Update on multidrug resistance efflux pumps in *Acinetobacter* spp. *Antimicrob Agents Chemother* 2021;**65**:e00514–21.

Kumar P, Lee JH, Lee J. Diverse roles of microbial indole compounds in eukaryotic systems. *Biol Rev* 2021;**96**:2522–45.

Kyriakidis I, Vasileiou E, Pana ZD *et al.* *Acinetobacter baumannii* antibiotic resistance mechanisms. *Pathogens* 2021;**10**:373.

Lau YY, How KY, Yin WF *et al.* Functional characterization of quorum sensing LuxR-type transcriptional regulator, EasR in *Enterobacter asburiae* strain L1. *PeerJ* 2020;**8:** e10068.

Lee JH, Cho MH, Lee J. 3-Indolylacetonitrile decreases *Escherichia coli* O157: H7 biofilm formation and *Pseudomonas aeruginosa* virulence. *Environ Microbiol* 2011;**13**:62–73.

Lee JH, Wood TK, Lee J. Roles of indole as an interspecies and interkingdom signaling molecule. *Trends Microbiol* 2015;**23**:707–18.

Lin F, Xu Y, Chang Y *et al.* Molecular characterization of reduced susceptibility to biocides in clinical isolates of *Acinetobacter baumannii*. *Front Microbiol* 2017;**8**:1836.

López M, Blasco L, Gato E *et al.* Response to bile salts in clinical strains of *Acinetobacter baumannii* lacking the AdeABC efflux pump: virulence associated with quorum sensing. *Front Cell Infect Microbiol* 2017;**7**:143.

Lv B, Zeng Y, Zhang H. Mechanosensitive channels mediate hypoionic shock-induced aminoglycoside potentiation against bacterial persisters by enhancing antibiotic uptake. *Antimicrob Agents Chemother* 2021;**66**:e0112521. AAC-01125.

Magnet S, Courvalin P, Lambert T. Resistance-nodulation-cell division-type efflux pump involved in aminoglycoside resistance in *Acinetobacter baumannii* strain BM4454. *Antimicrob Agents Chemother* 2001;**45**:3375–80.

Mishra R, Panda AK, De Mandal S *et al.* Natural anti-biofilm agents: strategies to control biofilm-forming pathogens. *Front Microbiol* 2020;**11**:2640.

Nguyen M, Joshi SG. Carbapenem resistance in *Acinetobacter baumannii*, and their importance in hospital-acquired infections: a scientific review. *J Appl Microbiol* 2021;**131**:2715–38.

Nowak J, Zander E, Stefanik D *et al.* High incidence of pandrug-resistant *Acinetobacter baumannii* isolates collected from patients with ventilator-associated pneumonia in Greece, Italy and Spain as part of the MagicBullet clinical trial. *J Antimicrob Chemother* 2017;**72**:3277–82.

Pfaffl MW. Relative quantification. *Real-time PCR* 2006;**63**:63–82.

Piperaki ET, Tzouvelekis LS, Miriagou V *et al.* Carbapenem-resistant *Acinetobacter baumannii*: in pursuit of an effective treatment. *Clin Microbiol Infect* 2019;**25**:951–7.

Pu Y, Ke Y, Bai F. Active efflux in dormant bacterial cells–new insights into antibiotic persistence. *Drug Resist Updat* 2017;**30**: 7–14.

Rangel K, Chagas TP, De-Simone SG. *Acinetobacter baumannii* infections in times of COVID-19 pandemic. *Pathogens* 2021;**10**: 1006.

Raorane CJ, Lee JH, Lee J. Rapid killing and biofilm inhibition of multidrug-resistant *Acinetobacter baumannii* strains and other microbes by iodoindoles. *Biomolecules* 2020;**10**:1186.

Ridone P, Vassalli M, Martinac B. Piezo1 mechanosensitive channels: what are they and why are they important. *Biophys Rev* 2019;**11**:795–805.

Roy S, Chatterjee S, Bhattacharjee A *et al.* Overexpression of efflux pumps, mutations in the pumps' regulators, chromosomal mutations and AAC6'Ib-cr are associated with fluoroquinolone resistance in diverse sequence types (STs) of neonatal septicaemic *Acinetobacter baumannii*: a 7-year single centre study. *Front Microbiol* 2021;**12:** 202.

Saipriya K, Swathi CH, Ratnakar KS *et al.* Quorum-sensing system in *Acinetobacter baumannii*: a potential target for new drug development. *J Appl Microbiol* 2020;**128**:15–27.

Selvaraj A, Valliammai A, Sivasankar C *et al.* Antibiofilm and antivirulence efficacy of myrtenol enhances the antibiotic susceptibility of *Acinetobacter baumannii*. *Sci Rep* 2020;**10**:1–4.

Sethupathy S, Sathiyamoorthi E, Kim YG *et al.* Antibiofilm and antivirulence properties of indoles against *Serratia marcescens*. *Front Microbiol* 2020;**584812**:2637.

Sidhu H, Capalash N. Synergistic anti-cancer action of salicylic acid and cisplatin on HeLa cells elucidated by network pharmacology and in vitro analysis. *Life Sci* 2021;**282**:119802.

Şimşek M, Demir C. Determination of colistin and tigecycline resistance profile of *Acinetobacter baumannii* strains from different clinical samples in a territory hospital in Turkey. *J Health Sci Med Res* 2020;**38**:81–91.

Tacconelli E, Carrara E, Savoldi A *et al.* Discovery, research, and development of new antibiotics: the WHO priority list of antibiotic-resistant bacteria and tuberculosis. *Lancet Infect Dis* 2018;**18**:318–27.

Uruén C, Chopo-Escuin G, Tommassen J *et al.* Biofilms as promoters of bacterial antibiotic resistance and tolerance. *Antibiotics* 2020;**10**:3.

Vázquez-López R, Solano-Gálvez SG, Juárez Vignon-Whaley JJ *et al.* *Acinetobacter baumannii* resistance: a real challenge for clinicians. *Antibiotics* 2020;**9**:205.

Vijay S, Bansal N, Rao BK *et al.* Secondary infections in hospitalized COVID-19 patients: Indian experience. *Infect Drug Resist* 2021;**14**:1893.

Vijayakumar S, Anandan S, Prabaa D *et al.* Insertion sequences and sequence types profile of clinical isolates of carbapenem-resistant *A . baumannii* collected across India over four year period. *J Infect Publ Health* 2020;**13**:1022–8.

Wood TK. Combatting bacterial persister cells. *Biotechnol Bioeng* 2016;**113**:476–83.

Zeighami H, Valadkhani F, Shapouri R *et al.* Virulence characteristics of multidrug resistant biofilm forming *Acinetobacter baumannii* isolated from intensive care unit patients. *BMC Infect Dis* 2019;**19**:1–9.

Zhao X, Zhao L, Zhao Y *et al.* 3-Indoleacetonitrile is highly effective in treating influenza a virus infection in vitro and in vivo. *Viruses* 2021;**13**:1433.

Zhu J, Li Q, Li X *et al.* Successful control of the first carbapenem-resistant *Klebsiella pneumoniae* outbreak in a Chinese hospital 2017–2019. *Antimicrob Resist Infect Cont* 2020;**9**:1–9.

Zhu LJ, Chen XY, Hou PF. Mutation of CarO participates in drug resistance in imipenem-resistant *Acinetobacter baumannii*. *J Clin Lab Anal* 2019;**33**:e22976.